WHITE PAPER · HWACHEON HI-ECO 21 HS × INDUSTRIAL & GENERAL MANUFACTURING

# Hwacheon Hi-Eco 21 HS for Industrial & General Manufacturing

High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most. This paper covers what the machine does, the industrial & general manufacturing parts we produce with it, the alloys we run, the tolerances and finishes industrial & general manufacturing buyers expect, and the documentation package.

**8" OD × 12" L max turning**
ENVELOPE

**±0.0005" routine on critical features**
TOLERANCE

**1994**
SINCE

**Same-week / rig-down direct**
RESPONSE




**Bearkat Blur** · 8" OD × 12" L max turning · high-speed precision CNC turning center

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## Executive summary

**Machine:** Hwacheon Hi-Eco 21 HS (shop nickname: *Bearkat Blur*). high-speed precision CNC turning center at B&R Productions, running from our New Waverly, Texas shop.

**Application:** High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most for Industrial & General Manufacturing customers — Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

**Envelope:** 8" OD × 12" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

## The machine, in context

Bearkat Blur exists because certain oilfield and aerospace parts are small, high-quantity, and can't tolerate the setup time of a larger machine. Bushings, couplings, precision shafts, valve inserts, and any repeatable geometry in the 8-inch envelope — this machine cycles fast without giving up accuracy. When cycle time drives the quote, this is the machine that wins.

## Why Bearkat Blur for Industrial work

Small industrial turned parts in volume — bushings, spacers, pins, small couplings, seat rings — where the whole quote turns on price per piece. 8" × 12", 2" bar through the spindle, fast rapids and a 12-station turret is a configuration built to win exactly that quote.

### What this looks like on a real part

Bar-fed bushing work is where the 2-inch through-spindle capacity does the real work. Feeding from bar removes a handling step per part, and at a few hundred pieces that step is most of your labour cost. The machining is undemanding; the economics are entirely about how few times a human touches the part between raw bar and finished piece.

> **When this is the wrong machine**
>
> 8-inch hard ceiling and 15 HP continuous. Bigger industrial turned parts go to a Hi-Eco 35 cell; forged or scaled stock goes to Hammer of Forge.

## Full specifications

| | |
|---|---|
| **Model** | Hwacheon Hi-Eco 21 HS ("Bearkat Blur") |
| **Type** | High-speed CNC turning center — small-part specialist |
| **Max turning capacity** | 8" OD × 12" L (B&R working spec) |
| **Platform max swing** | 22.8" over bed; 11.02" turning diameter platform-max |
| **Distance between centers** | 15.74" platform |
| **Bar capacity** | 2" (through-spindle) |
| **Chuck size** | 8" |
| **Spindle nose** | A2-6; 2.4" spindle bore |
| **Spindle speed** | 4,000 RPM max |
| **Spindle power** | 15 HP continuous / 20 HP 30-min rating |
| **Turret** | 12-station |
| **Rapid traverse** | 1,181 IPM on both X and Z axes |
| **Control** | Fanuc 18-T |
| **Machine weight** | ~8,900 lb |
| **Tolerance capability** | ±0.0005" routine on critical features |
| **Materials** | Stainless, brass, tool steel, Inconel 718/625, 17-4 PH, Monel |

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

## Who this serves in Industrial & General Manufacturing

**Typical buyer:** Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

**What's hard about industrial & general manufacturing work:** Legacy parts with no OEM support, one-off replacement components, small-lot production runs, materials that don't fit standard shop capabilities, and tight lead times on repair work where a down-machine is losing money by the day.

## Typical Industrial & General Manufacturing parts we produce on Bearkat Blur

- Precision pump components (impellers, sleeves, seal areas)
- Small bushings, sleeves, and bearing components
- Precision rollers and small shafts
- Custom-fit replacement parts for legacy equipment
- High-quantity industrial small parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

## Alloys we run for Industrial & General Manufacturing

**Standard range:** Stainless (304, 316, 17-4 PH) · Tool steels · 4140 / 4340 · Brass · Bronze · Aluminum · Cast iron · Specialty alloys per customer requirement.

Industrial work covers the widest alloy range because the applications are the widest. Food and pharma often specify 316L or 17-4 PH; mining and heavy equipment specify tool steels and 4140/4340; pump manufacturers outside oilfield use everything from bronze bushings to stainless impellers. We work with the customer's spec sheet — if you tell us the alloy, we'll tell you honestly whether we can machine it and how quickly.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

## How we machine it — our 5-step process

This is the process B&R runs on every job, industrial & general manufacturing or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

## Tolerance and finish expectations in Industrial & General Manufacturing

±0.0005" routine on critical features. Industrial applications range from precision (pump seal areas, close-fit bushings) to structural (mining and heavy-equipment brackets); we adjust the inspection plan to what the part actually needs. Surface finish measured on parts where finish drives function.

## Documentation and quality workflow

Documentation calibrated to customer requirement. Material traceability standard for critical parts. Certificates of conformance issued with delivery. NDAs on request for proprietary designs or reverse-engineered legacy parts.

## Response time and emergency work

Down-machine and production-critical work prioritized. Lead times honestly quoted — no false promises. For legacy reverse-engineering, we typically confirm feasibility on the first phone call. Call (936) 291-7827.

## How we compare

Compared to sourcing a new OEM part: we're often faster and cheaper for one-offs and small runs, and for legacy parts where the OEM has moved on we're often the only realistic option. Compared to a cheapest-bidder job shop: we hold tolerance, document material, and don't ruin the part.

## Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

## Frequently asked questions

8 questions covering the Hwacheon Hi-Eco 21 HS platform and Industrial & General Manufacturing work. Also indexed as FAQ schema for AI answer engines.

### What's the max turning diameter on the Hi-Eco 21 HS?

8" OD × 12" L is B&R's working spec. The platform itself has a 22.8" swing over bed and can accept turning diameter up to 11.02" with 15.74" between centers. Bar capacity: 2" through spindle.

### How fast is this machine — what's the spindle RPM?

4,000 RPM max spindle speed. Rapid traverse 1,181 IPM on both X and Z. Combined with the 8" chuck this is our high-cycle-count small-part specialist.

### What's the turret capacity?

12-station turret. Enough tool positions for most single-setup precision turning jobs including grooving, threading, and finish operations.

### What kind of industrial customers does B&R work with?

Industrial equipment OEMs, custom machinery builders, non-oilfield pump and valve manufacturers, food and pharma equipment suppliers, mining and heavy-equipment operators, and any customer who needs a legacy or reverse-engineered part made right.

### Do you machine one-off replacement parts?

Yes — a lot of our industrial work is exactly this. Send the sample or drawing; if the material is on the shelf, we're often faster than the OEM. Reverse-engineered legacy parts across almost every sector.

### What materials do you run for industrial work?

Full range — stainless (304, 316, 17-4 PH), tool steels, 4140/4340, brass, bronze, aluminum, cast iron, and specialty alloys per customer requirement. If you need something unusual, ask.

### Do you offer emergency turnaround on down-machine parts?

Yes for repeat customers with material on the shelf. Same-day and next-day realistic when we already have the alloy. Call (936) 291-7827 and describe what's down.

### Can you do small-batch production or is B&R just one-offs?

Both. Prototype and small-batch through repeat production. Small production runs where a big shop's minimum order would kill the economics are exactly where we fit.

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## Ready to talk about your Industrial & General Manufacturing work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

**Published by B&R Productions** — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.

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